```
0010  REM THIS PROGRAM CALCULATES THE SUGAR AND MOLASSES IN STOCK IN CUBIC FEET CALLED "MRSTOCKC"
0020  DIM A$(18)32
0030  FOR I=1 TO 18
0040  FOR J=1 TO 12
0050  E(I,J)=0
0060  NEXT J
0070  NEXT I
0080  A$(1)="Juice"
0090  A$(2)="Syrup"
0100  A$(3)="A Massecuite"
0110  A$(4)="Top-Off A Mass."
0120  A$(5)="B Massecuite"
0130  A$(6)="C Massecuite"
0140  A$(7)="A Molasses"
0150  A$(8)="Top-Off A Mol."
0160  A$(9)="B Molasses"
0170  A$(10)="Grain"
0180  A$(11)="B Magma"
0190  A$(12)="C Magma"
0200  A$(13)="Other"
0210  A$(14)="TOTAL"
0220  A$(15)="Final Molasses Purity"
0230  A$(16)="Sugar Purity"
0240  A$(17)="Sugar in Stock"
0250  A$(18)="Final Molasses in Stock"
0260  INPUT "1   JUICE, CU. FT.",E(1,1)
0270  INPUT "2   JUICE, BRIX",E(1,2)
0280  INPUT "3   JUICE, POL",E(1,3)
0290  INPUT "4   SYRUP, CU. FT.",E(2,1)
0300  INPUT "5   SYRUP, BRIX",E(2,2)
0310  INPUT "6   SYRUP, POL",E(2,3)
0320  INPUT "7   A MASSECUITE, CU. FT.",E(3,1)
0330  INPUT "8   A MASSECUITE, BRIX",E(3,2)
0340  INPUT "9   A MASSECUITE, POL",E(3,3)
0350  INPUT "10  TOP-OFF A MASSECUITE, CU. FT.",E(4,1)
0360  INPUT "11  TOP-OFF A MASSECUITE, BRIX",E(4,2)
0370  INPUT "12  TOP-OFF A MASSECUITE, POL",E(4,3)
0380  INPUT "13  B MASSECUITE, CU. FT.",E(5,1)
0390  INPUT "14  B MASSECUITE, BRIX",E(5,2)
0400  INPUT "15  B MASSECUITE, POL",E(5,3)
0410  INPUT "16  C MASSECUITE, CU. FT.",E(6,1)
0420  INPUT "17  C MASSECUITE, BRIX",E(6,2)
0430  INPUT "18  C MASSECUITE, POL",E(6,3)
0440  INPUT "19  A MOLASSES, CU. FT.",E(7,1)
0450  INPUT "20  A MOLASSES, BRIX.",E(7,2)
0460  INPUT "21  A MOLASSES, POL.",E(7,3)
0470  INPUT "22  TOP-OFF A MOLASSES, CU. FT.",E(8,1)
0480  INPUT "23  TOP-OFF A MOLASSES, BRIX",E(8,2)
0490  INPUT "24  TOP-OFF A MOLASSES, POL",E(8,3)
0500  INPUT "25  B MOLASSES, CU. FT.",E(9,1)
0510  INPUT "26  B MOLASSES, BRIX",E(9,2)
0520  INPUT "27  B MOLASSES, POL",E(9,3)
0530  INPUT "28  GRAIN, CU. FT.",E(10,1)
0540  INPUT "29  GRAIN, BRIX",E(10,2)
0550  INPUT "30  GRAIN, POL",E(10,3)
0560  REM INPUT "31  B MAGMA, CU. FT.",E(11,1)
0570  REM INPUT "32  B MAGMA, BRIX",E(11,2)
0580  REM INPUT "33  B MAGMA, POL",E(11,3)
0590  INPUT "34  C MAGMA, CU. FT.",E(12,1)
0600  INPUT "35  C MAGMA, BRIX",E(12,2)
0610  INPUT "36  C MAGMA, POL",E(12,3)
```

```
0620  INPUT "37  OTHER MATERIAL, CU. FT.",E(13,1)
0630  INPUT "38  OTHER MATERIAL, BRIX",E(13,2)
0640  INPUT "39  OTHER MATERIAL, POL",E(13,3)
0650  INPUT "40  FINAL MOLASSES PURITY FOR RUN",E(15,1)
0660  INPUT "41  SUGAR PURITY FOR RUN",E(16,1)
0670  FOR J=1 TO 13
0680  IF E(J,2)=0 THEN 740
0690  E(J,4)=E(J,3)*100/E(J,2)
0700  E(J,5)=62.2511+.24081*E(J,2)+.0007902404*E(J,2)^2+.00000423954*E(J,2)^3-1.657193E-8*E(J,2)^4
0710  E(J,6)=E(J,1)*E(J,5)/2000
0720  E(J,7)=E(J,6)*E(J,2)/100
0730  E(J,8)=E(J,6)*E(J,3)/100
0740  NEXT J
0750  FOR J=1 TO 8
0760  E(14,J)=E(1,J)+E(2,J)+E(3,J)+E(4,J)+E(5,J)+E(6,J)+E(7,J)+E(8,J)+E(9,J)+E(10,J)+E(11,J)+E(12,J)+E(13,J)
0770  NEXT J
0780  E(14,2)=E(14,7)*100/E(14,6)
0790  E(14,3)=E(14,8)*100/E(14,6)
0800  E(14,4)=E(14,3)*100/E(14,2)
0810  E(14,5)=E(14,6)*2000/E(14,1)
0820  E(17,1)=E(14,7)*(E(14,4)-E(15,1))/(E(16,1)-E(15,1))
0830  E(17,2)=E(17,1)*E(16,1)/100
0840  E(18,1)=E(14,7)-E(17,1)
0850  E(18,2)=E(14,8)-E(17,2)
0860  E(17,3)=E(17,2)*2000/.96
0870  K1=62.2511+.24081*H(30)+.0007902404*H(30)^2+.00000423954*H(30)^3-1.657193E-8*H(30)^4
0880  K1=K1*H(30)/748
0890  E(18,3)=E(18,1)*2000/K1
0900  SELECT PRINT 215(132)
0902 IF H(36)=0 THEN 910
0904 SELECT PRINT 204(132)
0910  IF H(23)=0 THEN 940
0920 PRINT HEX(0E);TAB(33-LEN(B$)/2);B$
0930 GOTO 950
0940 PRINT TAB(66-LEN(B$)/2);B$
0950  PRINT HEX(0A0A)
0955 IF H(23)=0 THEN 968
0960  PRINT HEX(0E);TAB(24);"STOCK CALCULATIONS"
0965 GOTO 970
0968  PRINT TAB(57);"STOCK CALCULATIONS"
0970  PRINT HEX(0A0A)
0980  PRINT "Date: ";A1$
0990  PRINT "--------------"
1010  PRINT HEX(0A0A)
1020 GOTO 1040
1040  PRINT "Material in Stock     Cu. Ft. Measured    Brix      Pol     Purity    Pounds per Cu. Ft.    Tons Material     Tons Brix
    Tons Pol"
1050  PRINT "-----------------     ---------------    -----    -----    ------    ------------------    -------------     ---------
    --------"
1060  FOR J=1 TO 13
1070  PRINTUSING 1080,A$(J),E(J,1),E(J,2),E(J,3),E(J,4),E(J,5),E(J,6),E(J,7),E(J,8)
1080  %################   -###,###,###.##     -###.##  -###.##   -###.##      -#,###.###      -##,###,###.### -#,###,###.### -###,#
##.###
1090  NEXT J
1100  PRINT "-------------           ----------       -----    -----    ------          ------              --------        -------
     -------"
1110  J=14
1120  PRINTUSING 1080,A$(J),E(J,1),E(J,2),E(J,3),E(J,4),E(J,5),E(J,6),E(J,7),E(J,8)
1130  PRINT "-------------           ----------       -----    -----    ------          ------              --------        -------
     -------"
1140  PRINT HEX(0A0A0A0A0A)
```

```
1150  PRINT "Assumptions"
1160  PRINT "-----------"
1170  PRINT HEX(0A)
1180  FOR J=15 TO 16
1190  PRINTUSING 1200,A$(J),E(J,1)
1200  %################################      -###.##
1210  NEXT J
1220  PRINT HEX(0A0A0A0A0A)
1230  PRINT "                                                    SUGAR AND MOLASSES IN PROCESS"
1240  PRINT "                                                    ------------------------------"
1250  PRINT HEX(0A0A)
1260  PRINT "                                      Lbs Brix           Lbs Pol"
1270  PRINT "                                      --------           -------"
1280  PRINTUSING 1290,A$(17),E(17,1)*2000,E(17,2)*2000,E(17,3)
1290  %########################   -###,###,###,###  -###,###,###,###        (Lbs. 96 Sugar in Stock = -##,###,###.##)
1300  PRINT HEX(0A)
1310  PRINTUSING 1320,A$(18),E(18,1)*2000,E(18,2)*2000,H(30),E(18,3)
1320  %########################   -###,###,###,###  -###,###,###,###        (Final Molasses in Stock - Gals. @ ##.## Brix =-#,###,##
#.##)
1330  PRINT "                                     ---------          --------"
1340  PRINTUSING 1350,E(14,7)*2000,E(14,8)*2000
1350  %TOTAL                      -###,###,###,###  -###,###,###,###
1360  PRINT "                                     ---------          --------"
1370  PRINT HEX(0C)
1380  SELECT PRINT 005(64)
1390  INPUT "IS DATA OKAY (Y OR N)",J$
1400  IF J$="Y" THEN 1430
1410  IF J$="N" THEN 260
1420  GOTO 1390
1430  LOAD T "MRSTART" 10
```